ELCT 361

Solutions for H/W #2

Q1

(a) The coordinates of M and N are used to write the two position vectors $\vec{A}$ and $\vec{B}$ in Fig.(a)

$$\vec{A} = x_1 \hat{a}_x + y_1 \hat{a}_y + z_1 \hat{a}_z$$

$$\vec{B} = x_2 \hat{a}_x + y_2 \hat{a}_y + z_2 \hat{a}_z$$

$$\vec{B} - \vec{A} = (x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z$$

(b) $$\vec{A} = (0-2)\hat{a}_x + [-2-(-4)]\hat{a}_y + (0-1)\hat{a}_z = -2\hat{a}_x + 2\hat{a}_y - \hat{a}_z$$

$$|A| = \sqrt{(-2)^2 + (2)^2 + (-1)^2} = 3$$

Therefore,

$$\hat{a}_A = \frac{\vec{A}}{|A|} = \frac{1}{3}(-2\hat{a}_x + 2\hat{a}_y - \hat{a}_z)$$

Q2. From the discussions we made in the class (Refer to Lecture chapter 2-2-7),

The differential surface element is $dS_r = r\,d\phi\,dz$, therefore.

$$A = \int_0^h \int_{\phi_1}^{\phi_2} r\,d\phi\,dz = \int_0^h \int_{\phi_1}^{\phi_2} dS_r$$

Where, $r = 2$, $h = 5m$, $30° \le \phi \le 120°$

Then, $$A = \int_0^5 \int_{\frac{\pi}{6}}^{\frac{2\pi}{3}} 2\,d\phi\,dz = 5\pi\,[m^2]$$

(b) Referring to our lecture materials, "chapter 2-2-9"

$$x = r\cos\phi, \quad y = r\sin\phi, \quad r = \sqrt{x^2+y^2}$$

Therfore,

$$\vec{A} = r\sin\phi\,\hat{a}_x + r\cos\phi\,\hat{a}_y + r\cos^2\phi\,\hat{a}_z$$

In addition, Refering to Lecture presentation materials, chapter 2-2-8,

$\vec{A}\cdot\hat{a}_r$, $\vec{A}\cdot\hat{a}_\phi$, (Z axis shared by Cartesian & cylindrical coordinates)

$$\hat{a}_x\cdot\hat{a}_r = \cos\phi \qquad \hat{a}_x\cdot\hat{a}_\phi = -\sin\phi \qquad \hat{a}_x\cdot\hat{a}_z = 0$$

$$\hat{a}_y\cdot\hat{a}_r = \sin\phi \qquad \hat{a}_y\cdot\hat{a}_\phi = \cos\phi \qquad \hat{a}_y\cdot\hat{a}_z = 0$$

$$\hat{a}_x\cdot\hat{a}_{\cancel{x}} = 0 \qquad \hat{a}_z\cdot\hat{a}_\phi = 0 \qquad \hat{a}_z\cdot\hat{a}_z = 1$$

Therefore,

$$\hat{a}_x = \cos\phi\,\hat{a}_r - \sin\phi\,\hat{a}_\phi$$

$$\hat{a}_y = \sin\phi\,\hat{a}_r + \cos\phi\,\hat{a}_\phi$$

$$\hat{a}_z = \hat{a}_z$$

Then,

$$\vec{A}\,(\text{in cylindrical}) = 2r\sin\phi\cos\phi\,\hat{a}_r + (r\cos^2\phi - r\sin^2\phi)\,\hat{a}_\phi + r\cos^2\phi\,\hat{a}_z$$

Q3

(a) Total charge Q will be defined by ; (from volume charge, ρ_v)

$$Q=\int_V \rho_v dv$$

Therefore, in spherical coordinates, $0 \le \phi \le 2\pi$, $0 \le \theta \le \pi$ $R=2cm$,

$$Q=\int_V \rho_v dv = \int_0^{2\pi}\int_0^{\pi}\int_{R=0}^{2\times 10^{-2}} 4\cos^2\theta \, dv$$

dv in spherical coordinate,

$$dv = \cancel{dR}\, R d\phi \, R\sin\theta d\theta \, dR = R^2 \sin\theta \, dR \, d\theta \, d\phi$$

$$Q=\int_0^{2\pi}\int_0^{\pi}\int_0^{2\times 10^{-2}} 4\cos^2\theta \, R^2 \sin\theta \, dR\, d\theta\, d\phi$$

$$= 4\int_0^{2\pi}\int_0^{\pi}\left(\frac{R^3}{3}\right)\Bigg|_0^{2\times 10^{-2}} \sin\theta\cos^2\theta \, d\theta\, d\phi$$

$$= \left(\frac{32}{3}\times 10^{-6}\right)\int_0^{2\pi}\left(-\frac{\cos^3\theta}{3}\right)\Bigg|_0^{\pi} d\phi$$

$$= \frac{64}{9}\times 10^{-6}\int_0^{2\pi} d\phi \cong \frac{128\pi}{9}\times 10^{-6} \cong 44.68\ [\mu C]$$

(b) Using the transformation for A_R, A_θ, A_ϕ we discussed in our lecture notes, Chapter 2-2-18,

$$A_R = A_x \sin\theta\cos\phi + A_y \sin\theta\sin\phi + A_z\cos\theta$$
$$= (x+y)\sin\theta\cos\phi + (y-x)\sin\theta\sin\phi + z\cos\theta.$$

Using the expression for x, y, and z

$$A_R = (R\sin\theta\cos\phi + R\sin\theta\sin\phi)\sin\theta\cos\phi + (R\sin\theta\sin\phi - R\sin\theta\cos\phi)\sin\theta\sin\phi + R\cos^2\theta$$

$$= R\sin^2\theta(\cos^2\phi + \sin^2\phi) + R\cos^2\theta$$

$$= R\sin^2\theta + R\cos^2\theta = R$$

similarly,

$$A_\theta = (x+y)\cos\theta\cos\phi + (y-x)\cos\theta\sin\phi - z\sin\theta$$

$$A_\phi = -(x+y)\sin\phi + (y-x)\cos\phi,$$

following the same processed with finding A_R, then, we obtain

$$A_\theta = 0, \quad A_\phi = -R\sin\theta.$$

Therefore,

$$\vec{A}(\text{in spherical}) = A_R\hat{a}_R + A_\theta\hat{a}_\theta + A_\phi\hat{a}_\phi = R\hat{a}_R - R\sin\theta\,\hat{a}_\phi$$

Q4.

(a) $V = x^2y + xy^2 + xz^2$

$$\nabla V = \left(\frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z\right)\cdot(x^2y + xy^2 + xz^2)$$

$$= (2xy + y^2 + z^2)\hat{a}_x + (x^2 + 2xy)\hat{a}_y + 2xz\,\hat{a}_z$$

Then,

$$\nabla V\Big|_{(1,-1,2)} = 3\hat{a}_x + (1-2)\hat{a}_y + 4\hat{a}_z$$

$$= 3\hat{a}_x - \hat{a}_y + 4\hat{a}_z$$

(b) The function E is expressed in terms of cylindrical variables. Hence, we need to below equation

$$\nabla E = \left(\frac{\partial}{\partial r}\hat{a_r} + \frac{1}{r}\frac{\partial}{\partial \phi}\hat{a_\phi} + \frac{\partial}{\partial z}\hat{a_z}\right)\left(E_0 e^{-2r}\sin 3\phi\right)$$

$$= -2E_0 e^{-2r}\sin 3\phi\,\hat{a_r} + \frac{(3E_0 e^{-2r}\cos 3\phi)}{r}\hat{a_\phi}$$

$$= \left[-2\sin 3\phi\,\hat{a_r} + \frac{3\cos 3\phi}{r}\hat{a_\phi}\right]E_0 e^{-2r} \qquad \boxed{\nabla E\Big|_{(1,\frac{\pi}{2},3)} = 0.27E_0\hat{a_r}}$$

Q5.

(a) $\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \quad (\vec{E} = E_x\hat{a_x} + E_y\hat{a_y} + E_z\hat{a_z})\ (\nabla = \frac{\partial}{\partial x}\hat{a_x} + \frac{\partial}{\partial y}\hat{a_y} + \frac{\partial}{\partial z}\hat{a_z})$

$$= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(x^2 z)$$

$$= 6x + x^2$$

$$\boxed{\nabla \cdot \vec{E}\Big|_{(2,-2,0)} = 16}$$

(b) $\nabla \cdot \vec{E}$ at spherical coordinates,

$$\nabla \cdot \vec{E} = \frac{1}{R^2}\frac{\partial}{\partial R}(R^2 E_R) + \frac{1}{R\sin\theta}\frac{\partial}{\partial \theta}(E_\theta \sin\theta) + \frac{1}{R\sin\theta}\frac{\partial E_\phi}{\partial \phi}$$

$$= \frac{1}{R^2}\frac{\partial}{\partial R}(a^3\cos\theta) + \frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}\left(-\frac{a^3\sin^2\theta}{R^2}\right)$$

$$= 0 - \frac{2a^3\cos\theta}{R^3} = -\frac{2a^3\cos\theta}{R^3}$$

$$\boxed{\nabla \cdot \vec{E}\Big|_{(\frac{a}{2},0,\pi)} = -16}$$